

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T04 · Powers, Roots and Standard Form

Questions

Student book · questions with writing space

■ 33 pages

FAMILY MAP

Powers Roots Standard Form

F01	Convert and compare standard form	14 Q 25 marks
F02	Calculate and model in standard form	12 Q 28 marks
F03	Simplify numerical index expressions	6 Q 10 marks

2 marks

(2)

Question 08

4MA1-1H Q15

MODULAR UNIT 1

3 marks

15 (a) $\sqrt{2} \div \frac{8^3}{16^2} = 2^n$

Work out the value of n
Show your working clearly.

$n = \dots\dots\dots$
(3)

1 (a) Simplify $e^8 \div e^2$

$$m = \dots\dots\dots (1)$$

$n =$

$p =$

8 (a) Write 520 million in standard form.

.....
(1)

(b) Write 8.79×10^{-5} as an ordinary number.

.....
(1)

- (c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

.....
(2)

Question 05

4MA1-1H Q9

MODULAR UNIT 2

3 marks

- 9 The table gives the amount of rice produced by each of two countries in 2020

Country	Amount of rice (tonnes)
Indonesia	3.5×10^7
Argentina	8.2×10^5

- (a) Write 3.5×10^7 as an ordinary number.

.....
(1)

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

- (b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

..... tonnes
(2)

Question 06

4MA1-1H Q9 (a)

MODULAR UNIT 2

1 marks

- 9 (a) Write 5.87×10^{-4} as an ordinary number.

.....
(1)

Question 07

4MA1-1H Q9 (b)

MODULAR UNIT 2

1 marks

(b) Write 84 000 000 in standard form.

.....
(1)

Question 08

4MA1-1H Q9 (c)

MODULAR UNIT 2

2 marks

The number of neurons in a human brain is 8.5×10^{10}
The number of neurons in a monkey brain is 1.47×10^9

The number of neurons in a human brain is $K \times$ the number of neurons in a monkey brain.

- (c) Work out the value of K
Give your answer correct to one decimal place.

$K = \dots\dots\dots$
(2)

Question 13

4MA1-1H Q15

MODULAR UNIT 2

3 marks

- (b) Find 4% of 4.5×10^{157}
Give your answer in standard form.

.....

(3)

Question 20

4MA1-2H Q6

MODULAR UNIT 2

2 marks

6 (a) Write 76 000 000 in standard form.

.....
(1)

(b) Write 5.4×10^{-4} as an ordinary number.

.....
(1)

Question 08

4MA1-1H Q9 M01

MODULAR UNIT 2

1 marks

- 9 (a) Write 8.4×10^{-5} as an ordinary number.

.....

(1)

Question 09

4MA1-1H Q9 M02

MODULAR UNIT 2

2 marks

- (b) Work out $(6.5 \times 10^{-40}) \times (8 \times 10^{185})$
Give your answer in standard form.

.....
(2)

Question 09

4MA1-1H Q8 Q8(a)

MODULAR UNIT 2

1 marks

8 (a) Write 0.000089 in standard form.

.....
(1)

Question 10

4MA1-1H Q8 Q8(b)

MODULAR UNIT 2

1 marks

(b) Write 8.34×10^4 as an ordinary number.

.....
(1)

4MA1-1H Q8

2 marks

$$b = 3 \times 10^{145}$$

Give your answer in standard form.

4MA1-2H Q12

3 marks

Give your answer in standard form.

Give your answer in standard form.

(b) Work out $\frac{9.6 \times 10^{141} + 6.4 \times 10^{140}}{3.2 \times 10^{16}}$

Give your answer in standard form.

Country	Population
China	1.4×10^9
Germany	8.2×10^7
Sweden	9.9×10^6
Fiji	9.1×10^5
Malta	4.3×10^5

- (a) Work out the difference between the population of China and the population of Germany. Give your answer in standard form.

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

- $$k = \dots\dots\dots (2)$$

- 9 A rainwater tank contains 2.4×10^7 raindrops.
The rainwater tank also contains 1.75×10^6 bacteria.

- (a) Work out the number of bacteria per raindrop in the tank.
Give your answer in standard form correct to 2 significant figures.

.....
(3)

A drop of rainwater contains 5.01×10^{21} atoms.

In a drop of rainwater the number of atoms is 3 times the number of molecules.

- (b) Work out the number of molecules in the rainwater tank.
Give your answer in standard form correct to one significant figure.

..... molecules
(2)

(3)

..... molecules
(2)

(1)

(1)

(2)

(Total for Question 5 is 4 marks)

9 (a) Write 6.25×10^{-4} as an ordinary number.

.....
(1)

(b) Work out $(2.4 \times 10^{12}) \div (9.6 \times 10^4)$
Give your answer in standard form.

.....
(2)

(Total for Question 9 is 3 marks)

(1)

(2)

(Total for Question 8 is 3 marks)